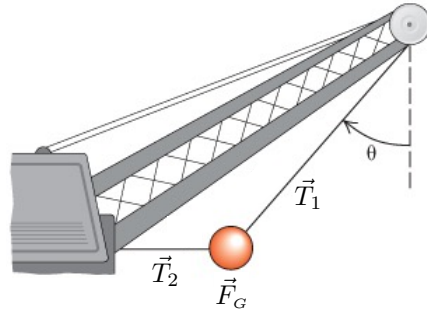


Practice: Newton's First Law

Sketch your FBDs according to Handout 05.1, and follow the solution steps according to Handout 05.3.

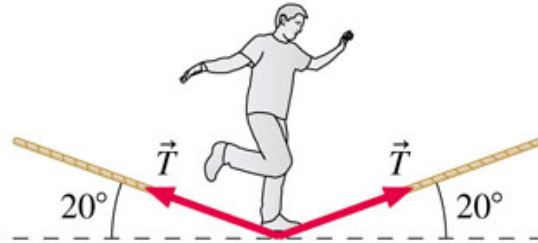
Problem: A wrecking ball weighing 2500 N hangs from a first cable. Prior to swinging, it is pulled back to an angle $\theta = 20^\circ$ by a second horizontal cable. The wrecking ball is in equilibrium before release.



What are the tensions in the first and second cables?

[Ans: $T_1 = 2660$ N, $T_2 = 910$ N]

Problem: A student with a mass of 62 kg (weight of 608 N) is walking on a slackline, a length of webbing stretched between two trees. The line stretches and so has a noticeable sag as shown in the figure. At the point where his foot touches the line, the rope applies a tension force in each direction as shown.



What is the tension in the line?

[Ans: 889 N]